

INTERNSHIP REPORT

AI-Powered Business Automation Assistant Using LLM Integration and Machine Learning

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Internship at ADVLST | 25 May 2026 – 10 July 2026

About the Organization

ADVLSST

*IT Solutions & AI Automation Company
New Delhi, India*

- Software Infrastructure
- Backend Systems
- Automation Solutions
- AI Research & Development Division

ROLE

AI & Machine Learning Intern

DEPARTMENT

AI Research & Development

DURATION

25 May 2026 – 10 July 2026

MODE

Remote / Work From Home

Introduction & Problem Statement

Enterprises handle large volumes of repetitive queries, data-processing tasks and communication workflows — rule-based automation is rigid and struggles with natural-language variability and unstructured data.

Goal: Build an AI-driven assistant that interprets natural-language requests, applies ML models for structured prediction, and uses LLM integration for accurate, context-aware responses supporting business-automation workflows.

Key Objectives

- Understand ML & LLMs in business automation
- Design and preprocess required datasets
- Build and evaluate ML models
- Integrate LLMs via APIs
- Apply prompt-engineering techniques
- Evaluate outputs with appropriate metrics

Technologies & Tools Used



Python

Main programming language



Pandas / NumPy

Data handling & numerical ops



Scikit-learn

Machine-learning implementation



LLM APIs

Natural-language understanding & generation



Prompt Engineering

Reliable, structured LLM outputs



REST APIs

Model, data & app-layer integration

System Architecture



Structured requests route to the trained ML model; natural-language requests are converted into an engineered prompt and sent to the LLM API. Both paths converge at the response-generation layer.

Methodology

1

Requirement Understanding

2

Data Collection & Inspection

3

Cleaning & Preprocessing

4

Feature Engineering

5

Model Training & Evaluation

6

Prompt Design & LLM Integration

7

End-to-End Testing

8

Error Analysis & Refinement

9

Documentation

Model Evaluation Measures

Measure	Purpose
Accuracy	Overall proportion of correct predictions
Precision	Reliability of positive predictions
Recall	Ability to identify relevant positive cases
F1-score	Combined measure of precision and recall
Response Relevance (LLM)	Qualitative assessment of LLM output usefulness

Expected Outcomes & Learnings

Working prototype combining ML models and LLM-based NLP

Structured data pipeline for preprocessing & feature handling

Hands-on experience with prompt engineering & LLM API integration

Trained and evaluated ML model(s) for the chosen task

Understanding of end-to-end AI-automation workflow design

Foundation for future enhancement (fine-tuning, scaling)

Certificate of Internship

Ref: ADV-INT-2026-152



CERTIFICATE OF INTERNSHIP

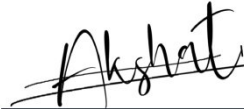
THIS DOCUMENT IS PROUDLY PRESENTED TO

Rounak Gupta

In recognition of his successful completion of the **AI & Machine Learning** internship program at ADVLIST Strategy.
During his tenure, Rounak demonstrated exceptional analytical skills, technical proficiency in developing machine learning models, and a strong commitment to delivering high-quality AI-driven solutions.

We deeply appreciate his dedication and wish him continued success in all future professional endeavors.

For verification mail us at
work@advlst.in


CHIEF DIRECTOR

Conclusion

This internship provided practical, industry-level exposure to Artificial Intelligence and Machine Learning, with a specific focus on integrating Large Language Models into business-automation workflows — combining classical ML with modern generative-AI capabilities to build real-world solutions.

Thank You

Rounak Gupta | AI & Machine Learning Intern, ADVLST